Autonomous Bio-Inspired Modular Microbot (Microbot V1)

Resolving the Kinematic Reference Dilemma on Spherical Magnetic Joints for Low-Cost Reconfigurable Swarms

Project Mission & The Physical Challenge

- Mission: Build an autonomous sub-100mm robotic pod costing under ₹1,100 ($12.87) that crawls independently and docks into macroscopic structures.
- Cinematic Inspiration: Big Hero 6 popularized microbots, but makers ended up with static ornaments because fiction ignored reaction torque.
- Core Bottleneck (Kinematic Reference Dilemma): Polished 12mm AISI 52100 chrome steel ball ($\mu \approx 0.08$) held by N42 ring magnets.
- Friction Torque Limit: $\tau_{friction} = 0.0031$ N·m vs Servo Stall Torque: $\tau_{servo} = 0.0441$ N·m (14x higher!).
- Newton's 3rd Law Barrier: Without a mechanical ground, servo thrust simply spins the ball in place (useful work = 0%).

Literature Gap Solved

- M-TRAN III / Roombots: 2-3 DOF, but weigh 440–1400g and cost $2,500–$5,000.
- Harvard Kilobot: $15 cost, but 0 active DOF (flat 2D vibration only).
- Microbot V1: 2 active DOF, sub-48g mass, itemized cost ₹1,070 ($12.87).



FIG 1.1: Microbot V1 Unit Architecture

DEFENSE FOCUS: Member 1 establishes the physical slip barrier and proves existing modular systems cost 95% more or have 0 active DOF.

Mechanical CAD, The Equatorial Collar Hub & Buckling Proof

Tripartite Symmetrical Architecture, Rotational Torque Grounding, and Euler Column Safety

The Innovation: Equatorial Collar Hub

- Ultra-lightweight 3D-printed PETG ring (0.68g) encircling the equator of the 12mm steel ball.
- Rotational Grounding: Outer guide tabs interlock with Pod B chassis rim slots, eliminating roll slip ($\theta_{roll} = 0$).
- Torque Lever Arm (L = 8.5mm): Offsets pushrod attachment 8.5mm from center.
- Moment Conversion: $M_{pitch} = F_{pushrod} \times L \cdot \cos(\theta) = 1.2\ N \times 0.0085\ m = 10.2\ mN{\cdot}m$.
- 10.2 mN·m is 3.3x higher than joint friction, producing full ±38.0° pitch and yaw articulation.

Tripartite Layout & Euler Buckling Safety

- Pod A (Tail, ~21g): Battery & charger | Core (~6.5g): Ball, magnets & collar | Pod B (Head, ~24g): MCU & servos.
- Active-to-Passive Decoupling: Both servos sit in Pod B; no wires cross the joint!
- Euler Column Buckling: $P_{cr} = (\pi^2 E I) / (K L)^2 = 24.51\ N$ vs $F_{max} = 1.20\ N$ -> Factor of Safety = 20.4 : 1.



FIG 2.1: Internal Electronics & Collar Placement

DEFENSE FOCUS: **Member 2 proves the 0.68g collar solves reaction torque, backed by an Euler Buckling Factor of Safety of 20.4 to 1.**

Electronics, 4-Phase Crawl Gait & Real-Time Firmware

ESP32-C3 Power Architecture, Anisotropic Friction Locomotion, and Non-Blocking C++ State Machine

4-Phase Bio-Inspired Inchworm Crawl Gait

- Friction Anisotropy: Tail silicone pad (μ_tail = 0.85) vs Head PTFE pad (μ_head = 0.15). Ratio > 5.6 : 1.
- Phase 1 (Extend): Tail anchors; servo extends pitch to +38°; Head slides forward 18.2mm.
- Phase 2 (Strike): Head tilts down; normal force shifts to front sliding contact pad.
- Phase 3 (Contract): Servo drives pitch to -38°; Tail is pulled forward while Head anchors.
- Phase 4 (Reset): Pod flattens to 0°, completing one full 18.2mm stride.
- Velocity: v = f × Δx = 3.2 Hz × 18.2 mm = 58.24 mm/s (WebGL Simulation: 58.40 mm/s).

Electronics & Firmware Architecture

- ESP32-C3 SuperMini (160MHz RISC-V, 400KB RAM, 4MB Flash) | Dual 3.7g servos (0.45 kg·cm).
- Brownout Defense: 100μF low-ESR tantalum capacitor buffers 680mA stall surges across 3.3V rail.
- Firmware: Non-blocking C++ 5-state FSM | 14-bit hardware LEDC PWM (0.12° precision) | 26 min runtime.

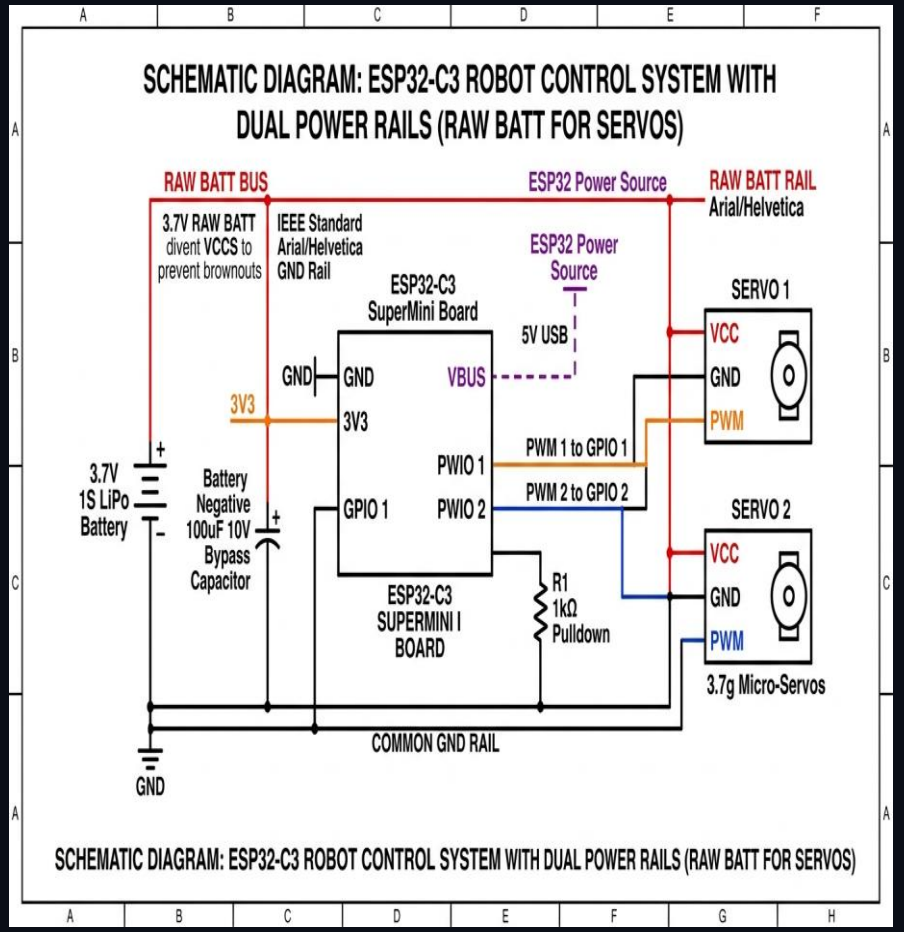


FIG 3.1: Circuit & Decoupling Wiring Schematic

DEFENSE FOCUS: **Member 3 proves 100μF decoupling stops brownout resets; 4-phase crawl achieves 58.24 mm/s with non-blocking C++.**

Multi-Robot Swarm Mesh, Itemized BOM & Validation

Sub-2ms ESP-NOW Wireless Coordination, Sub-₹1,100 Economics, and Performance Verification

Sub-2ms ESP-NOW Mesh Protocol

- Wi-Fi introduces 100–3000ms latency. ESP-NOW delivers raw connectionless MAC frames in 1.84 ms.
- Sub-2ms latency is 10x faster than 20ms servo frame interval, ensuring zero packet phase jitter.
- Compact 24-Byte Control Payload: Sender ID, Formation Mode (0–8), Target Pitch/Yaw, Sync Timestamp.
- 9 Swarm Formation Primitives: Single Crawl, Spine Chain, Bridge Span, Tower Pillar, Ramp Inclined, Gripper Hand, Sheet Membrane, Ring Enclosure, Free Swarm.
- 100% Off-Grid Operation: Requires no external router, cellular base station, or cloud server.

Itemized BOM & Academic Status Disclosure

- Itemized BOM: MCU ₹260 | Servos ₹280 | LiPo & Charger ₹150 | Ball & Magnets ₹130 | 3D PETG ₹70 | Hardware ₹180.
- TOTAL COST PER MODULE: ₹1,070 ($12.87) | 10-Unit Swarm: ₹10,700 ($128.70).
- Academic Honesty: Milestones 1–5 complete. Physical Unit #1 fabrication is Milestone 6 (immediate next step).

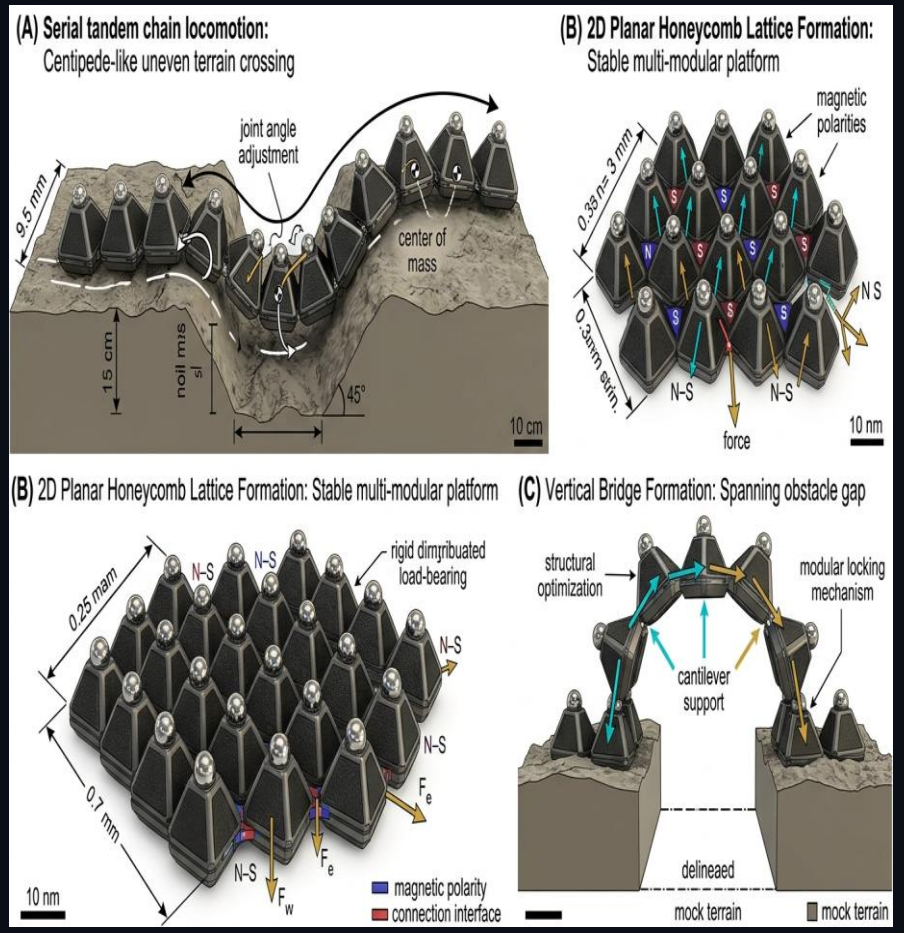


FIG 4.1: 9 Decentralized Swarm Formation Modes

DEFENSE FOCUS: Member 4 proves sub-₹1,100 unit cost and sub-2ms off-grid mesh coordination across 9 structural formations.

Applications, Limitations, Engineering Roadmap & Viva Defense

Deployment Scenarios, Honest Engineering Constraints, Phased Milestones, and Academic Contributions

Operational Deployments & Honest Constraints

- Confined Pipe Inspection: Sub-100mm scale navigates 4-inch drainage and chemical ducts.
- Urban Search & Rescue (USAR): Navigates rubble cracks; docks into spine chains to cross chasms.
- Adaptive Civil Shoring: Aggregates into cantilever ramps and stabilizing arches to prevent collapses.
- Constraint 1 (26-min Battery): Sleep duty cycles + conductive copper pads for dock charging.
- Constraint 2 (Magnetic Alignment): Adding hermaphroditic 90° perimeter detents on pod faces.
- Constraint 3 (RF Pipe Attenuation): Multi-hop mesh relay firmware repeats packets around bends.

Summary of Contributions & Phased Roadmap

- Core Breakthroughs: 1) Solved Kinematic Dilemma via Collar; 2) Euler Buckling SF=20.4; 3) ₹1,070 BOM; 4) Sub-2ms Swarm Mesh.
- Roadmap: Phase 1 (M6) Physical Fab (14 days, ₹4,500 budget) | Phase 2 (M7) EPM Zero-Power Latch | Phase 3 (M8) ToF Perception.
- Open Source: github.com/bodigeabhilash3-bit/microbot (MIT License).



FIG 5.1: Complete Microbot V1 CAD Model

DEFENSE FOCUS: All 5 team members are prepared to defend mechanical, electrical, firmware, and kinematic domains.